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Electric Motor

Abstract

Various embodiments of the teachings herein include an electric motor. An example includes: a stator having a plurality of bar-shaped field conductors; a plurality of inverters for actuating the field conductors; and a cooling apparatus including a fluid pipe externally adjacently guided on one or more of the field conductors. The inverters are arranged on one or more printed circuit boards arranged on at least one cooling plate. The at least one cooling plate is mechanically connected to the field conductors or conductors electrically connected to the field conductors.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a U.S. National Stage Application of International Application No. PCT/EP2023/058318 filed Mar. 30, 2023, which designates the United States of America, and claims priority to EP application Ser. No. 22/170,036.2 filed Apr. 26, 2022, the contents of which are hereby incorporated by reference in their entirety.

TECHNICAL FIELD

[0002] The present disclosure relates to electric motors. Various embodiments of the teachings herein include an electric motor with a bar winding on the stator side.

BACKGROUND

[0003] Electric motors can have a bar winding on the stator side. Here, the stator (stator) has a series of bars instead of wound wire conductors as field conductors. The bars have a low inductance compared to conventional windings. Therefore, a comparatively high current flow is necessary for generating a predefined magnetic field.

[0004] However, this high current flow requires only a comparatively low voltage of, for example, 12 v due to the low resistance of the bars. The low voltage makes it possible to arrange the components of the inverters, with which the bars are actuated, at small spacings from one another. The components of the power electronics can thus be arranged, for example, on one or more printed circuit board(s) which are arranged close to the electric motor. The bars can be used directly or via electrically conductive bar-like connecting elements directly as mechanical supports for the printed circuit boards.

[0005] The high heat loss of the electrical and electronic components, which is caused by the high current intensities, is therefore compounded by a high packing density of the electrical and electronic components in an electric motor configured in this way. The direct connection thereof to the stator/rotor block via bars provides for further heat input from the stator/rotor block. This disadvantageously results in a considerable cooling effort for the electrical and electronic components on the printed circuit board.

SUMMARY

[0006] Teachings of the present disclosure include an electric motors which avoid the disadvantage mentioned in the introduction. For example, some embodiments include an electric motor (10) with: a stator (11) with a plurality of field conductors (12) embodied as bars (12), a plurality of inverters for actuating the field conductors (12), wherein the inverters are arranged on one or more printed circuit board(s) (15), the printed circuit boards (15) are arranged on at least one cooling plate (16), the cooling plate (16) is arranged such that the field conductors (12) or conductors (18) electrically connected to the field conductors (12) are mechanically operatively connected to the cooling plate, at least one cooling apparatus is present which comprises a fluid pipe (20, 60) which is externally adjacently guided on one or more of the field conductor(s) (12).

[0007] In some embodiments, the fluid pipes (20, 60) are water pipes.

[0008] In some embodiments, a pump for a cooling fluid is present.

[0009] In some embodiments, a fluid pipe (20, 60) is guided on each of the field conductors (12)

and/or conductors (18).

[0010] In some embodiments, a region (19), in which the fluid pipe (20, 60) is externally adjacently guided on the field conductor (12) and/or conductor (18), is axially located between a cooling plate (16) and an axial beginning of the stator/rotor block (8).

[0011] In some embodiments, field conductor (12) and conductor (18) are connected by means of a shoe (17) and wherein the fluid pipe (20, 60) is guided along the shoe (17).

[0012] In some embodiments, the fluid pipes (20, 60) are connected together in series for at least two of the field conductors (12) or conductors (18).

[0013] In some embodiments, the fluid pipe (20, 60) encircles the ring, which the field conductors (12) or the conductors (18) form, at its outer edge or on the inside at its inner edge.

[0014] In some embodiments, the fluid pipe (20, 60) is guided along a radially outwardly or inwardly pointing side of the field conductors (12) or conductors (18) and has surface elements (61) adjacent to an azimuthal side of the field conductors (12) or conductors (18).

[0015] In some embodiments, the fluid pipe (20, 60) runs, at least in certain sections, parallel to the field conductor (12) or conductor (18).

[0016] In some embodiments, the fluid pipe (20, 60) has a ceramic material, in particular is made from a ceramic material.

[0017] In some embodiments, the fluid pipe (20, 60) is insulated from the field conductor (12) or conductor (18) by an electrically insulating layer of material.

[0018] In some embodiments, the printed circuit boards (15) are circular or ring sector-shaped.

[0019] In some embodiments, the cooling plate (16) is arranged perpendicular to the axis (9) of the electric motor (10).

[0020] In some embodiments, the electric motor (10) actuates each of the field conductors (12) with a separate phase.

[0021] In some embodiments, the inverters are embodied to generate an alternating voltage with an amplitude of 200 v or less, in particular 150 v or less, in particular 50 v or less.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0022] The teachings of the present disclosure are described and explained in more detail below with reference to the exemplary embodiments shown in the figures. In the drawings, schematically:

[0023] FIG. 1 shows an electric motor incorporating teachings of the present disclosure with three cooling plates for heat dissipation from printed circuit boards in an oblique view;

[0024] FIG. 2 shows printed circuit boards incorporating teachings of the present disclosure arranged on a cooling plate in a front view;

[0025] FIG. 3 shows a sectional image of the cooling plates with conductor bars and connecting elements;

[0026] FIG. 4 shows a side view of conductor bars and cooling plates with an embodiment of fluid channels;

[0027] FIG. 5 shows a non-perspective pattern for a series connection of the fluid channels;

[0028] FIG. 6 shows a further embodiment for a fluid channel with azimuthal guiding incorporating teachings of the present disclosure.

DETAILED DESCRIPTION

[0029] The electric motors described herein comprise a stator with a plurality of field conductors embodied as bars as well as plurality of inverters for actuating the field conductors. The inverters are arranged on one or more printed circuit board(s). Furthermore, the printed circuit boards are arranged on at least one cooling plate and the cooling plate is arranged in such a way that the field conductors or conductors electrically connected to the field conductors are mechanically

operatively connected to the cooling plate, in particular penetrate it. Finally, at least one cooling apparatus is present which comprises a fluid pipe which is externally adjacently guided on one or more of the field conductor(s).

[0030] Due to the fluid cooling, the electric motors described herein have improved heat dissipation for the field conductors. As a result, a smaller heat flow reaches the cooling plates and the electronic devices arranged on the cooling plates, in particular the inverters. As a result, the cooling effort in the region of the cooling plates and the electronic devices can in turn be reduced.

[0031] In some embodiments, the field conductors or conductors act as mechanical supports for the cooling plates and are therefore mechanically operatively connected to them. It is possible that the field conductors or conductors penetrate the cooling plate and the printed circuit board. In addition to solid bars, the field conductors embodied as bars can also be those which are composed of a plurality of sub-conductors. The sub-conductors can be connected in parallel if the field conductors are, for example, Roebel conductors. The sub-conductors can also be connected in series and the field conductors consist of a plurality of sub-components which are lined up, for example, per welded joint.

[0032] The conductors can be arranged to be mechanically fixed to the field conductors and be connected to them, with exactly one field conductor always being connected to exactly one conductor, with the conductors axially extending the field conductors. In this case, the field conductors, conductors and cooling plate form a mechanically rigid unit.

[0033] In some embodiments, the fluid pipes can be water pipes. Water is a standard coolant and can be provided, for example in a factory, from outside of the electric motor, so pipes suitable for water are expedient.

[0034] In some embodiments, the electric motor can have a pump for a cooling fluid. If the cooling fluid is not supplied from outside, a circulation of fluid can be generated which provides for heat removal due to movement of material. If the motor has a high number of field conductors, for example more than 50 or even more than 100 field conductors, a plurality of pumps can also be present.

[0035] In some embodiments, the electric motor can be configured such that a fluid pipe is guided on each of the field conductors and/or conductors. It is expedient for each of the field conductors or conductors to provide heat dissipation since during operation of the electric motor the field conductors or conductors are equally subjected to heat and thus each of the field conductors or conductors contributes approximately equally to the heat transport to the cooling plates.

[0036] In some embodiments, a region, in which the fluid pipe is externally adjacently guided on the field conductor and/or conductor, is axially located between a cooling plate and an axial beginning of the stator/rotor block. The field conductors and/or conductors lie freely in this region and the fluid pipe can be attached without problems. Furthermore, it is expedient to remove the heat in this region in order to reduce the heat input from the rotor region, i.e. the drive system in the electronic devices, i.e. the region of the cooling plates and printed circuit boards.

[0037] If conductors are present then these are expediently connected to the field conductors by means of a shoe, wherein the fluid pipe is guided along the shoe. The shoe serves as a permanent connection between the field conductors and the conductors, with the field conductors pertaining to the drive side of the electric motor and the conductors making contact with the electronic devices on the printed circuit boards and supporting the cooling plates and printed circuit boards. In some embodiments, the shoe can be geometrically adapted to the fluid pipe without the shape or the cross-section of the field conductors having to be changed. Production of the field conductors is simplified as a result. In some embodiments, the fluid pipe can be externally adjacently guided on a field conductor.

[0038] The fluid pipes for at least two of the field conductors or conductors can be connected together in series. In other words, a single fluid pipe leads to one field conductor or conductor and then in succession to at least one further field conductor or conductor. Only thereafter does the fluid

pipe lead away from the field conductors, for example to a reservoir. In this way a plurality of the field conductors or conductors are cooled by a single fluid pipe, and this reduces the connection effort.

[0039] The fluid pipe can encircle the ring, which the field conductors or the conductors form, at its outer edge or on the inside at its inner edge. In other words, the fluid pipe extends azimuthally beyond at least two of the field conductors or conductors. The fluid pipe can form a complete ring, i.e. touch all field conductors or conductors, or else form only a ring sector in which the fluid pipe extends over some of the field conductors or conductors and thus removes their heat. In the latter case it is expedient to use a plurality of such fluid pipes which together extend over and thus cool all field conductors or conductors. For this it is also expedient to use a fluid pipe for an angular segment or a portion of the field conductors or conductors, which corresponds to a fraction of the full circle with an integral denominator, i.e. $\frac{1}{3}$, $\frac{1}{4}$ or $\frac{1}{8}$ of the full circle. For example, in the case of 72 field conductors, $\frac{1}{6}$ of them, i.e. 12 could be cooled by one fluid pipe and six such fluid pipes could be present. Each of the fluid pipes would thus cover an angle of 60° of the full circle.

[0040] The fluid pipe can be guided along a radially outwardly or inwardly pointing side of the field conductors or conductors and has surface elements adjacent to an azimuthal side of the field conductors or conductors. The surface elements enlarge the contact area between the fluid pipe and the field conductor or conductor and thereby provide for improved heat transfer from field conductor or conductor to the fluid. If the fluid pipes are guided along a radially external side of the field conductors or conductors, the surface element expediently points radially inwards, and vice versa. The surface element can be U-shaped and thus can provide for mechanical retention of the fluid pipe in addition to better thermal contact.

[0041] The fluid pipe can run, at least in certain sections, parallel to the field conductor or conductor. In other words, in certain sections it is arranged on the field conductor or conductor such that the fluid channel runs parallel to the field conductor or conductor.

[0042] The fluid pipe may comprise a ceramic material, in particular be made from a ceramic material. Ceramic materials are heat-resistant, chemically stable and electrically insulating and therefore allow installation in the electric motor without requiring further measures such as electrical insulation.

[0043] The fluid pipe can be electrically insulated from the field conductor or conductor by an electrically insulating layer of material. Insulating pads are examples of this. It is particularly advantageous in this connection again that, owing to its construction, the electric motor operates with relatively low voltages and thus the effort for electrical insulation can be kept comparatively low. The thickness of the layer of material and an additional insulation directed toward the side only have to be adjusted to voltage differences in the region of, for example, 12 V.

[0044] The electric motor can have a plurality of printed circuit boards. In particular, a plurality of separate printed circuit boards can be provided on a cooling plate. The power electronics used can be modularized by way of a distribution of a plurality of printed circuit boards. By using a large number of similar printed circuit boards it is thus possible to provide a large number of converters, whereby production in respect of rejects is improved.

[0045] The printed circuit boards can be circular or ring sector-shaped. As a result, the printed circuit boards can be arranged in a circle or a ring which encloses the axle of the electric motor. Printed circuit boards with this shape may be assembled to form a circle or ring and thus, optimally adjusted to the shape of the electrical machine, be arranged at an axial end of the machine, with a high modularity simultaneously being achieved. As a result, the printed circuit boards can be compactly arranged axially offset relative to the stator and rotor and form an integral part of the electric motor. In this way it is also possible to arrange a plurality of cooling plates with printed circuit boards axially offset from one another and thus connect inverters, for example, in parallel which are arranged at the same azimuthal location.

[0046] In some embodiments, the cooling plate is arranged perpendicularly to the axis of the

electric motor. In this way the cooling plate with the printed circuit boards can be compactly arranged at an axial end of the electrical machine. A plurality of cooling plates can also be arranged axially offset and close to one another. Furthermore, with such an arrangement a uniform spacing between cooling plate and power electronics from the bars, which form the field conductors, is given, whereby establishing contact with the bars is simplified.

[0047] In some embodiments, each of the field conductors is actuated with a separate phase. A phase is taken to mean an alternating voltage supply which is phase-offset with respect to all other phases used in the electric motor by an angle that is different from zero. In this case, a separate inverter is expediently present for each of the field conductors and actuates only this field conductor.

[0048] The inverters can generate an alternating voltage with an amplitude of 200 v or less, in particular 150 v or less, in particular 50 v or less. The voltage generated in this way is the voltage to which the field conductors, i.e. the stator bars, are subjected. This comparatively low voltage makes it possible for the first time for the components of the inverters to be arranged very close to one another. Spacings of approximately 2 mm can be used between the components, such as the power semiconductor switches, resulting in a high packing density of the electronic components and the possibility of arranging a large number of inverters on a comparatively small space. As a result, in the case of a small space requirement it is again possible to use a high number of phases, in particular a number of phases which corresponds to the number of stator bars. With an appropriately high number of stator bars it is thus possible to use 48, 72 or even 120 phases.

[0049] The terms “axial”, “radial” and “azimuthal” refer here to the axis of the rotor and therewith to the corresponding axis of symmetry of the stator. Here, “axial” describes a direction parallel to this axis, “radial” describes a direction orthogonal to the axis, towards or away from it, and “azimuthal” is a direction which is directed at a constant radial distance from the axis and at a constant axial position circularly around the axis.

[0050] When the terms “axial”, “radial” and “tangential” are used with reference to a surface, for example a cross-sectional surface, the terms describe the orientation of the normal vector of the surface, i.e. of that vector which is perpendicular to the surface concerned.

[0051] FIG. 1 represents an isometric view of an electric motor **10** incorporating teachings of the present disclosure. The electric motor **10** comprises a stator **11** and a rotor which is arranged substantially in the stator **11** and is not visible in FIG. 1. The rotor is connected in a rotationally fixed manner to a shaft, which is likewise not represented in FIG. 1. By electromagnetic interaction of the rotor with a stator **11** energized with current, the rotor is set in rotation about an axis **9**. The rotor is separated from the stator **11** by an air gap.

[0052] In some embodiments, the electric motor **10** can also be an external rotor motor or bell-type armature motor.

[0053] The stator **11** comprises a plurality of rigid and straight conductor bars **12** as field conductors. On the end face **13** which is averted in FIG. 1, these conductor bars **12** are connected together by a short-circuit ring. On the rear side **14** of the electric motor **10**, the conductor bars **12** are individually fed by respectively associated inverter modules. Since, due to the conductor bars **12**, an electric motor **10** operated at low voltages is involved, the inverter modules together with other components of the electronic devices (DC/DC converters, rectifiers) can be arranged relatively close together on printed circuit boards **15**. In this example, the printed circuit boards are in the form of ring sectors and many individual printed circuit boards **15** together form an annular printed circuit board structure.

[0054] While it is assumed in the examples that the boards **15** support inverter modules, it is also possible that some of the boards **15** support rectifiers and DC/DC converters.

[0055] FIG. 2 shows a plan view of such a circuit board structure. The number of circuit boards shown in FIG. 2 is reduced and greatly simplified compared with the illustration in FIG. 1 for better clarity. The specific number of such circuit boards **15** depends on the specific embodiment of

the electric motor **10**, in particular the number of conductor bars **12**. Each of the printed circuit boards **15** comprises a plurality of semiconductor switches **422**.

[0056] Further, some of the printed circuit boards **15** or all of the printed circuit boards **15** can comprise driver circuits not represented in the figures and other electronic components such as capacitors. The semiconductor switches **422** are power semiconductors such as IGBTs, MOSFETs, or JFETs, and can additionally comprise diodes (not shown) depending on the interconnection. The semiconductor switches **422** are connected, for example, as half bridges. A capacitor (not shown) can constitute, for example, an intermediate circuit capacitor of the half bridges. The semiconductor switches **422** of a circuit board **15** can be assigned to a single phase or else to a plurality of phases.

[0057] The printed circuit boards **15** furthermore comprise contact points **421** to which the conductor bars **12** are connected. The printed circuit boards **15** are supported by disk-shaped cooling plates **16**, it being possible for the cooling plates **16** to be occupied on both sides with printed circuit boards **15** for a better use of space.

[0058] Since relatively high currents are necessary in the conductor bars in the electric motor **10**, a plurality of inverters may be connected in parallel for the energization thereof. This can be achieved, for example, by the six circuit board structures shown in FIG. **1** on three cooling plates **16** all being connected in the same way to the conductor bars **12** and thus being connected electrically in parallel. In this case, use is made of the fact that the conductor bars **12** or connecting elements to the conductor bars **12** penetrate the cooling plates **16** and therewith also the printed circuit boards **15** in the same way at the contact points or at least contact them in the case of the outermost cooling plate **16**.

[0059] FIG. **3** shows a sectional image of the electric motor **10** in an oblique view. It can be seen here that the connecting elements **18** mechanically support the three cooling plates **16** and penetrate them. The connecting elements **18** are connected to the conductor bars **12** by shoes **17**. The inverters, which are located on the printed circuit boards in the regions in which one of the connecting elements **18** penetrates a cooling plate **16**, are connected in parallel and together provide the current for the current bar **12**.

[0060] The stator/rotor block **8** of the electric motor **10** and the electronic devices on the printed circuit board generate heat. In addition to the electrical conduction, the conductor bars **12** and the connecting elements **18** also constitute a heat bridge between the two components. However, it is important, particularly on the electronic device side, that there is not an additional excessive input of heat since the electronic components must not greatly exceed a temperature of approximately 80° C. for long. Intensified cooling in the region of the printed circuit boards is expensive, however.

[0061] It may therefore be advantageous to provide cooling in the region **19** between the cooling plates **16** and the stator/rotor block **8**. FIG. **4** shows such a type of cooling at a conductor bar **12**. In the case of the cooling provided in FIG. **4**, a fluid channel **20** is attached to the conductor bar **12** such that the water, which is used here as the cooling fluid, is guided along the conductor bar **12** via a distance in the region of a few cm. The volume of heat, which can pass from the conductor bar **12** to the fluid channel **20** and therewith the water, can be adjusted over the length of this distance since a longer distance produces a larger contact surface and the latter a greater flow of energy.

[0062] The fluid channel **20** can be, for example, ceramic. In this case, it is rigid in the manner of ceramic materials. A hose connection **23** can therefore be present at both ends, into which connection a hose leads for continued water supply. An inlet **21** and an outlet **22** are implemented as a result. These hoses can be brought together at a different point in order to implement a cooling circuit. At this point the heat can be emitted to the surroundings via a heat exchanger or heat sink. It is possible that said elements are designed as part of the electric motor **10**. In some embodiments, it is also possible that the cooling circuit is already present, independently of the electric motor **10**, at the installation site of the electric motor **10** and the electric motor **10** is connected to it together

with other devices.

[0063] In some embodiments, the fluid channel can also be manufactured from other materials, also, for example, metallic materials. In this case, an electrical line from the conductor bar **12** to other elements should be avoided, for example by using plastics material hoses for supplying water.

[0064] Since all conductor bars **12** are heated equally by the side of the stator/rotor block **8** and are connected equally to the printed circuit boards **15**, it is expedient to provide a fluid channel **20** for each of the conductor bars **12**. In particular embodiments of the electric motor **10**, the number of conductor bars **12** is high, however, for example 48, 72 or even 120. In the latter case, there are thus 120 outlets **22** and 120 inlets **21** which have to be connected or brought together at a different point and have to be reached for a flow of water, for example by way of a pump system.

[0065] For particular embodiments of the electric motor **10** it is therefore advantageous to combine fluid channels **20** that are close together into groups and to connect them in series. FIG. 5 schematically represents such an embodiment. It is also assumed that the fluid channels **20** are arranged directly on the conductor bars **12**. However, in order to simplify the representation, FIG. 5 shows neither the conductor bars **12** nor the actually three-dimensional, annular arrangement of the fluid channels **20**, but instead represents the fluid channels **20** side by side in a simplified manner.

[0066] According to FIG. 5, three fluid channels **20** are combined to form a group and are connected in series in respect of the flow of water. If this grouping is applied in an electric motor **10** with 120 conductor bars **12** for all conductor bars **12**, 40 such groups result and therefore 40 inlets **21** and 40 outlets **22**, i.e. far fewer than before. By forming larger groups the effort in respect of guiding the cooling fluid can be reduced further as long as the cooling capacity for the respectively last conductor bars **12** remains sufficiently high in the series of fluid channels.

[0067] In the embodiment according to FIG. 5, the flow of water still runs, in certain sections, parallel to the orientation of the conductor bars **12**. In some embodiments, which is represented in FIG. 6, the orientation of the flow of water is rotated by 90° and the water is distanced azimuthally around the ring which is embodied by the conductor bars **12**. FIG. 6 shows the thus formed fluid channel **60**. The fluid channel **60** rests on four of the conductor bars **12** over its course. According to FIG. 6, this does not correspond to all conductor bars **12**, but only to some of the conductor bars **12**. A similar consideration can take place in the formation of the groups in FIG. 5. The more conductor bars **12** that extend over the fluid channel **60**, the simpler installation and connection are since the number of inlets **21** and outlets **22** reduces. On the other hand, the potential cooling capacity reduces, however. Furthermore, for assembly purposes it can be simpler in this case with a rigid fluid channel **60** if it does not form a complete ring, i.e. only extends over some of the conductor bars **12**.

[0068] Since in the fluid channel **60** in the exemplary embodiment in FIG. 6 the water does not flow along the longitudinal extension of the conductor bars **12**, the distance, which is available for the delivery of water, and therewith also the surface, is shorter than in the embodiments of FIGS. 4 and 5. To guarantee an adequate heat transfer the fluid channel **60** can have additional surface-expanding elements **61**. These appear tooth-shaped in FIG. 6 and are made of full material to enable good heat conduction. In other words, they do not constitute an additional volume for the cooling fluid but instead expand the surrounding material such that an enlarged surface which touches the conductor bars **12** results.

[0069] Electrical insulation is automatically provided when plastics material hoses are used for the transverse joint between the fluid channels. The fluid channel **60**, by contrast, can also be designed as an integral component. If it is then made from an insulating material, for example a ceramic, insulation is then still provided between the conductor bars **12**. If the fluid channel **60** is metallic, by contrast, then insulation is necessary to prevent a cross circuit of the conductor bars **12**. For this purpose insulation pads can be arranged between the fluid channel **60** and optionally the additional surface elements **61** and the conductor bar **12**.

[0070] In the preceding exemplary embodiments, the fluid channel **20**, **60** is always guided along

the surface of the conductor bars **12**. In all embodiments the fluid channel **20, 60** can be guided equally as well along the connecting elements **18** and heat thus dissipated from the connecting elements **18**.

[0071] A further possibility, which can be substituted in all exemplary embodiments, is to guide the fluid channel **20, 60** along the shoes **17** with which the connecting elements **18** and the conductor bars **12** are connected together. This is advantageous to the extent that the shoes are typically already embodied to connect the conductor bars **12** and connecting elements **18** which are different in terms of their cross-sectional shape and therefore have more complex shapes themselves than the conductor bars **12** or the connecting elements **18**. It is therefore possible to take account of and compensate for a reduction in the conductive cross-section, which result[s] due to holes, in the construction of the shoes **17** without problems. Both the conductor bars **12** and the connecting elements **18** can thereby have a uniform construction again and thereby be easier to produce.

LIST OF REFERENCE NUMERALS

[0072] **8** stator/rotor block [0073] **9** motor axis [0074] **10** electric motor [0075] **11** stator [0076] **12** conductor bars [0077] **13** end face [0078] **14** rear side [0079] **15** printed circuit board [0080] **16** cooling plate [0081] **17** shoe [0082] **18** connecting element [0083] **19** region [0084] **20, 60** fluid channel [0085] **21** inlet [0086] **22** outlet [0087] **23** hose connections [0088] **61** additional surface element [0089] **421** contact points [0090] **422** semiconductor switch

Claims

1. An electric motor comprising: a stator having a plurality of bar-shaped field conductors; a plurality of inverters for actuating the field conductors; wherein the inverters are arranged on one or more printed circuit boards arranged on at least one cooling plate; wherein the at least one cooling plate is mechanically connected to the field conductors or conductors electrically connected to the field conductors; and a cooling apparatus including a fluid pipe externally adjacently guided on one or more of the field conductors.
2. The electric motor as claimed in claim 1, wherein the fluid pipes comprise water pipes.
3. The electric motor as claimed in claim 1, further comprising a pump delivering a cooling fluid to the cooling apparatus.
4. The electric motor as claimed in claim 1, wherein a fluid pipe is guided on each of the field conductors and/or conductors.
5. The electric motor as claimed in claim 1, wherein a region in which the fluid pipe is externally adjacently guided on the field conductor and/or conductor is axially located between a cooling plate and an axial beginning of the stator/rotor block.
6. The electric motor as claimed in claim 1, wherein the field conductor and conductor are connected by a shoe and wherein the fluid pipe is guided along the shoe.
7. The electric motor as claimed in claim 1, wherein the fluid pipes are connected together in series for at least two of the field conductors or conductors.
8. The electric motor as claimed in claim 1, wherein the fluid pipe encircles a ring formed by the field conductors or the conductors form, at an outer edge or on the inside at an inner edge.
9. The electric motor as claimed claim 8, wherein the fluid pipe is guided along a radially outwardly or inwardly pointing side of the field conductors or conductors and has surface elements adjacent to an azimuthal side of the field conductors or conductors.
10. The electric motor as claimed in claim 1, wherein the fluid pipe runs, at least in certain sections, parallel to the field conductor or conductor.
11. The electric motor as claimed in claim 1, wherein the fluid pipe comprises a ceramic.
12. The electric motor as claimed in claim 1, wherein the fluid pipe is insulated from the field conductor or conductor by an electrically insulating layer of material.
13. The electric motor as claimed in claim 1, wherein the printed circuit boards are circular or ring

sector-shaped.

14. The electric motor as claimed in claim 1, wherein the cooling plate is arranged perpendicular to the axis of the electric motor.

15. The electric motor as claimed in claim 1, operable to actuate each of the field conductors with a separate phase.

16. The electric motor as claimed in claim 1, wherein the inverters generate an alternating voltage with an amplitude of 200 v or less.
